

Maximal Semantic Clique Enumeration in Vector-Attributed Graphs

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ABSTRACT

Maximal clique enumeration (MCE) is a fundamental primitive for discovering cohesive structures in graph data, but structural completeness alone cannot distinguish semantically coherent groups in vector-attributed graphs. Enforcing collective semantic agreement is challenging because a global average-similarity constraint is non-monotone: a feasible clique may have infeasible subsets, invalidating the pivot invariants that make classical MCE scalable. We formalize *maximal semantic clique* (MSC) enumeration under an average pairwise similarity threshold and first establish SEMBK, a correct pivot-free baseline. We then develop two exact algorithms. STRSUB decouples structural and semantic search, applying Tomita pivoting in a τ -agnostic structural phase and admissible top-edge and vertex-contribution bounds during exact subset enumeration. MONOSEMMCE specializes in selective thresholds. Its key property is that every feasible clique has a feasible parent obtained by deterministically deleting a minimum-contribution vertex. The resulting reverse search follows only unique-parent edges and generates each feasible state once without visited tables or materialized size layers. Experiments verify exact output equality on all tested configurations. Across eight datasets, STRSUB and MONOSEMMCE jointly complete all 24 common-grid workloads, whereas SEMBK reaches the 1,200-second limit on 15; at q_{99} , MONOSEMMCE is $2.17\times$ – $6.24\times$ faster than STRSUB across all eight datasets and reaches $5.62\times$ parallel speedup. Output studies further show that MSCs attain higher semantic purity and vertex coverage than pairwise-threshold mining across q_{50} – q_{99} .

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1 INTRODUCTION

Graphs with rich vertex attributes are ubiquitous in modern data management and analytics. Social networks associate users with profile embeddings, knowledge graphs attach entity descriptions to nodes, and biological networks annotate proteins with functional vectors. In all these settings, the attribute space carries semantic meaning that is largely orthogonal to the topological structure: two vertices may be structurally adjacent yet semantically distant,

and vice versa. Identifying cohesive subgraphs that are simultaneously structurally dense and semantically coherent is therefore a recurring need across domains.

The maximal clique is a classic structural model of cohesion: every pair of vertices is connected, guaranteeing the highest possible edge density. However, in vector-attributed graphs, a clique may group vertices that are heterogeneous in the attribute space. Conversely, attribute-based community detection methods (e.g., modularity optimization with attribute homogeneity) often sacrifice structural guarantees and produce loosely connected groups. This tension motivates a joint model that enforces both structural completeness and semantic coherence.

The MSC Problem. We study the problem of enumerating *maximal semantic cliques* (MSCs) in a vector-attributed graph. Each vertex carries a real-valued attribute vector; pairwise semantic similarity is measured by cosine similarity. Given a threshold τ , a clique is a τ -semantic clique if its average pairwise similarity is at least τ . An MSC is an inclusion-wise maximal τ -semantic clique: no strict superset is also a τ -semantic clique. The task is to enumerate *all* MSCs of the input graph.

This formulation has two salient features. First, the average constraint is a *global* property of the clique: it depends on all $\binom{|C|}{2}$ pairwise similarities, not on any single edge. This makes it fundamentally different from edge-level semantic constraints (e.g., requiring every edge to exceed a threshold), which are local and monotone. Second, the average is non-monotone over the subset lattice: adding a vertex can raise or lower the average, so a partial clique that currently violates the constraint may still grow into a valid MSC, while one that satisfies it may admit no feasible extension. This non-monotonicity invalidates the structural invariants that underpin the efficiency of classical maximal clique enumeration (MCE) algorithms.

Why Classical MCE Does Not Suffice. The Bron-Kerbosch (BK) algorithm and its modern variants achieve state-of-the-art performance for MCE through two key techniques: *pivot-based pruning* (Tomita et al.) and *degeneracy ordering* (Eppstein et al.). Pivot pruning relies on the fact that structural maximality is closed under supersets: if a vertex u is adjacent to the current pivot p , any maximal clique containing u also contains p , so u can be safely suppressed as an expansion root. Under the average constraint, this closure breaks: a clique containing u but not p may satisfy $\text{avgSim} \geq \tau$ while $\{p, u\}$ does not. Consequently, pivoting is not merely ineffective—it is *incorrect* for MSC enumeration. A direct BK adaptation must therefore forgo pivoting entirely and defer all semantic evaluation to the leaves, degenerating to exhaustive structural enumeration regardless of τ .

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Our Contributions. This work formalizes MSC enumeration under inclusion-wise maximality and establishes its computational hardness. It develops STRSUB, which separates structural MCE from semantic subset enumeration, and MONOSEMMCE, a canonical reverse search specialized for selective thresholds. Experiments verify exact output equality, characterize their complementary runtime and memory regimes, and measure the resulting pattern space over a fixed threshold grid.

Paper Organization. Section 2 formalizes the problem and establishes hardness. Section 3 presents the baseline SEMBK algorithm and explains why pivoting fails. Section 4 develops STRSUB and MONOSEMMCE. Section 5 describes shared implementation optimizations. Section 6 reports experimental results. Section 7 discusses related work, and Section 8 concludes.

2 PRELIMINARIES

2.1 Notations and Problem Definition

We study a *vector-attributed graph* $\mathcal{G} = (V, E, \mathbf{X})$, where V is the vertex set, $E \subseteq \binom{V}{2}$ is the edge set, and $\mathbf{X} = \{\mathbf{x}_v \in \mathbb{R}^d \mid v \in V\}$ assigns a d -dimensional attribute vector to each vertex.

DEFINITION 1 (SIMILARITY FUNCTION). A similarity function $\text{sim} : \mathbb{R}^d \times \mathbb{R}^d \rightarrow [-1, 1]$ is a symmetric, bounded function measuring the semantic closeness of two attribute vectors. We assume all attribute vectors are ℓ_2 -normalized, so that sim coincides with cosine similarity and its range is naturally $[-1, 1]$.

Throughout this paper we work with normalized vectors and precompute $\text{sim}(u, v)$ for all $(u, v) \in E$. The threshold $\tau \in [-1, 1]$ is drawn from the same range.

DEFINITION 2 (CLIQUE). A set $C \subseteq V$ is a clique in $\mathcal{G}$ if every pair of distinct vertices in C is connected by an edge, i.e., $\{u, v\} \in E$ for all $u \neq v \in C$.

Classical maximal clique enumeration (MCE) seeks cliques that cannot be extended by any additional vertex while preserving the clique property. In vector-attributed graphs, however, structural density alone is insufficient to characterize semantically coherent groups: a clique may consist of vertices that are structurally adjacent yet semantically heterogeneous. To capture both aspects jointly, we introduce two companion quantities.

DEFINITION 3 (PAIRWISE SIMILARITY SUM AND AVERAGE PAIRWISE SIMILARITY). For a clique C of size $k = |C| \geq 2$, the pairwise similarity sum is

$$\text{simSum}(C) = \sum_{\{u, v\} \subseteq C} \text{sim}(u, v),$$

and the average pairwise similarity is

$$\text{avgSim}(C) = \frac{\text{simSum}(C)}{\binom{k}{2}}.$$

DEFINITION 4 (SEMANTIC CLIQUE). Given a threshold $\tau \in [-1, 1]$, a clique C with $|C| \geq 2$ is a τ -semantic clique if $\text{avgSim}(C) \geq \tau$.

For algorithmic purposes, we equivalently track the *semantic surplus*

$$W(C) = \text{simSum}(C) - \tau \binom{k}{2},$$

which is non-negative if and only if $\text{avgSim}(C) \geq \tau$. Working with simSum and W avoids division at every recursive step; avgSim is used only when a human-readable quantity is required.

DEFINITION 5 (MSC). Given $\mathcal{G} = (V, E, \mathbf{X})$ and threshold $\tau \in [-1, 1]$, a τ -semantic clique $C \subseteq V$ is a maximal semantic clique (MSC) if there is no τ -semantic clique $C' \supsetneq C$.

This is inclusion-wise maximality, not merely the absence of a feasible single-vertex extension. The distinction matters because average feasibility is non-monotone. For example, let $\tau = 0.6$ and let four vertices form a clique with $\text{sim}(a, b) = 0.6$, $\text{sim}(a, u) = \text{sim}(b, u) = \text{sim}(a, v) = \text{sim}(b, v) = 0.5$, and $\text{sim}(u, v) = 1$. The feasible clique $\{a, b\}$ has no feasible single-vertex extension, since either three-vertex average is $1.6/3 < 0.6$, but the strict superset $\{a, b, u, v\}$ is feasible with average $3.6/6 = 0.6$. Consequently, algorithms may use the absence of a feasible single extension only to generate candidates; a global containment test is still required for MSC output.

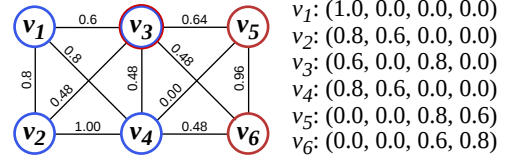


Figure 1: Running example with vector attributes and edge similarities.

EXAMPLE 1. Figure 1 shows a vector-attributed graph with six vertices. Edge labels give the precomputed pairwise similarities. With threshold $\tau = 0.6$, consider the two structural cliques

$$M_1 = \{v_1, v_2, v_3, v_4\}, \quad M_2 = \{v_3, v_4, v_5, v_6\}.$$

Let $C_1 = \{v_1, v_2, v_3, v_4\}$, $C_2 = \{v_3, v_5, v_6\}$, and $C_3 = \{v_3, v_4, v_5, v_6\}$. Their total and average similarities are

$$\begin{aligned} \text{simSum}(C_1) &= 4.16, & \text{avgSim}(C_1) &\approx 0.693 \geq \tau, \\ \text{simSum}(C_2) &= 2.08, & \text{avgSim}(C_2) &\approx 0.693 \geq \tau, \\ \text{simSum}(C_3) &= 2.56, & \text{avgSim}(C_3) &\approx 0.427 < \tau. \end{aligned}$$

Thus C_1 and C_2 are τ -semantic cliques, whereas the structurally valid clique C_3 is not.

Moreover, neither C_1 nor C_2 has a strict superset that is a τ -semantic clique, so both are MSCs.

Problem Statement.

PROBLEM 1 (MSC ENUMERATION). Given a vector-attributed graph $\mathcal{G} = (V, E, \mathbf{X})$ and a threshold $\tau \in [-1, 1]$, enumerate all MSCs of $\mathcal{G}$.

2.2 Challenges

Hardness. MSC enumeration generalizes classical MCE: when τ is at most the minimum pairwise similarity over all edges, every structural maximal clique satisfies the average constraint, so MSC enumeration coincides with MCE and the $\Omega(3^{n/3})$ lower bound carries over. The associated decision problem is NP-complete.

THEOREM 2.1. *Given $\mathcal{G} = (V, E, X)$, threshold $\tau \in [-1, 1]$, and integer k , deciding whether an MSC of size $\geq k$ exists is NP-complete.*

PROOF. *NP membership.* The decision instance is positive if and only if it contains a τ -semantic clique C of size at least k . The forward implication is immediate. For the reverse implication, the finite family of feasible clique supersets of C contains an inclusion-wise maximal member, whose size is at least $|C| \geq k$. A certificate therefore need only provide such a feasible clique C : verifying all edges and $\text{avgSim}(C) \geq \tau$ takes $O(|C|^2)$ time. It is unnecessary, and under the non-monotone constraint insufficient, to verify maximality by checking only single-vertex extensions.

NP-hardness. We reduce from the CLIQUE decision problem [24], which asks whether $G_0 = (V, E)$ contains a clique of size $\geq k$. Assign $\mathbf{x}_v = \mathbf{e}_1$ to every vertex, so $\text{sim}(u, v) = 1$ for every edge, and set $\tau = 1$. Under this assignment $\text{avgSim}(C) = 1 \geq \tau$ for every clique C , so every structural maximal clique is an MSC and vice versa. Any clique of size $\geq k$ in G_0 is contained in some structural maximal clique of size $\geq k$, which is also an MSC of size $\geq k$. Conversely, any MSC of size $\geq k$ is a clique of size $\geq k$ in G_0 . Hence a size- k MSC exists if and only if G_0 contains a clique of size $\geq k$, completing the reduction. $\square$

The output sets of MSC enumeration and MCE are incomparable: a structural maximal clique with $\text{avgSim}(M) < \tau$ may contain semantically maximal subsets that MCE never generates, while valid MSCs need not be structurally maximal. The output size, and hence the computational cost, is τ -dependent in a non-trivial way that we characterize below.

Output Distribution and Practical Scope. The number of MSCs and their sizes vary substantially with τ . At sufficiently low τ , almost every clique satisfies the average constraint and the output converges toward that of classical MCE. As τ increases, large structural cliques that previously satisfied the constraint cease to do so and are replaced by many smaller semantically feasible sub-cliques, none of which subsumes another. This *fragmentation effect* can inflate the output size by orders of magnitude relative to MCE. At sufficiently high τ , the constraint eliminates most cliques outright and the output shrinks considerably.

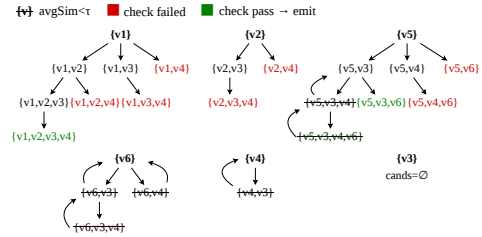
The intermediate fragmentation band carries the largest output volume and is therefore inherently time-consuming—any correct algorithm must enumerate all output cliques, so the output size itself imposes a lower bound on runtime. Low τ is appropriate when the goal is to filter out structurally maximal cliques that are semantically too loose; high τ is appropriate when the goal is to surface only the most semantically coherent groups. The exact τ values at which these transitions occur depend on the similarity distribution of the underlying dataset and cannot be fixed a priori. Section 6 therefore evaluates a fixed grid of per-dataset edge-similarity quantiles rather than choosing absolute thresholds after observing runtimes.

Non-monotonicity and Framework Incompatibility. The core difficulty is that avgSim is non-monotone over the subset lattice. In classical MCE, structural feasibility is monotone and maximality is enforced implicitly by the $P = \emptyset$ invariant at every leaf, so the set of feasible extensions can be characterized solely by the current state of C . Under the average constraint, no such invariant exists: a partial clique with $\text{avgSim}(C) < \tau$ may still grow into a valid MSC,

while one with $\text{avgSim}(C) \geq \tau$ may admit no feasible extension. Furthermore, whether a candidate vertex can participate in a valid extension depends on its accumulated similarity to the current partial clique, which evolves throughout the recursion and cannot be captured by structural conditions alone. As a result, the pruning mechanisms based on structural monotonicity lose their foundation, and the pivot strategy no longer applies. Direct adaptations of BK are therefore forced to defer all semantic evaluation to the leaves, where τ appears only in validation rather than guiding the search itself.

3 FRAMEWORK

Framework Overview. We derive a baseline search procedure by adapting the Bron-Kerbosch (BK) recursion to the semantic-clique setting under the average constraint. The standard partition (P, X) remains useful for structural enumeration, but the semantic constraint is defined at the clique level: whether a partial clique can eventually satisfy the threshold depends on avgSim over all vertex pairs within the clique. As a consequence, the pivoting strategy, which relies on local structural properties, is no longer directly applicable. The baseline therefore retains the BK recursion and degeneracy ordering while excluding pivoting, and enforces semantic maximality via a secondary CHECK pass on X at each leaf.



Algorithm 1: Semantic BK with Check

```

1 Input:  $G = (V, E)$ , semantic vectors  $\{x_v\}$ , threshold  $\tau$ 
2 Output: All maximal semantic cliques  $C$  s.t.  $\text{avgSim}(C) \geq \tau$ 
3 Precompute  $\text{sim}(u, v)$  and degeneracy ordering  $\sigma$ ;
4 for each  $v \in \sigma$  in parallel do
5    $P \leftarrow \{u \in N(v) \mid \sigma(u) > \sigma(v)\};$ 
6    $X \leftarrow \{u \in N(v) \mid \sigma(u) < \sigma(v)\};$ 
7   Initialize  $\text{acc}[u] \leftarrow \text{sim}(u, v)$  for all  $u \in N(v)$ ;
8   Search( $C = \{v\}, \text{simSum} = 0, P, X, \text{acc}, \text{mode} = \emptyset, 0$ )

9 Function Search( $C, \text{simSum}, P, X, \text{acc}, \text{mode}, k_{\min}$ ):
10   $k \leftarrow |C|$ ,  $\text{foundLonger} \leftarrow \text{false}$ ;
11  for each  $v \in P$  do
12     $\text{simSum}' \leftarrow \text{simSum} + \text{acc}[v]$ ;
13     $P \leftarrow P \setminus \{v\}, P' \leftarrow P \cap N(v), X' \leftarrow X \cap N(v)$ ;
14     $\text{acc}[u] \leftarrow \text{acc}[u] + \text{sim}(u, v)$  for  $u \in P' \cup X'$ ;
15     $\text{res} \leftarrow \text{Search}(C \cup \{v\}, \text{simSum}', P', X', \text{acc}, \text{mode}, k_{\min})$ ;
16     $\text{acc}[u] \leftarrow \text{acc}[u] - \text{sim}(u, v)$  for  $u \in P' \cup X'$ ;
17    if  $\text{res}$  then
18       $\text{foundLonger} \leftarrow \text{true}$ ;
19      if  $\text{mode} = \text{CHECK}$  then return true;
20     $X \leftarrow X \cup \{v\}$ ;
21  if  $\neg \text{foundLonger}$  then
22    if  $k \geq 2 \wedge 2 \cdot \text{simSum} / (k(k-1)) < \tau$  then return false;
23    if  $\text{mode} = \text{CHECK}$  then return  $k > k_{\min}$ ;
24    if  $X \neq \emptyset$  then
25      if Search( $C, \text{simSum}, X, \emptyset, \text{acc}, \text{CHECK}, k$ ) then
26        return true;
27    output  $C$ ;
28    return true;
29  return foundLonger;

```

mode with $k_{\min} = |C|$. In CHECK mode, the recursion explores extensions through X and returns *true* as soon as it finds any valid clique strictly larger than C ; if no such extension exists, C is emitted. The strict size condition $k > k_{\min}$ is necessary because the CHECK recursion may reach sub-cliques of C itself; without recording $k_{\min}$, these would be misidentified as valid extensions.

Why pivoting is not used. In standard BK, a pivot $p \in P \cup X$ is selected so that every maximal clique in the current branch contains either p or a vertex in $P \setminus N(p)$. Vertices in $P \cap N(p)$ are therefore never used as expansion roots: any maximal clique containing such a vertex u must also contain p and will be discovered in the sub-tree rooted at p . This argument is purely structural and relies on the fact that structural maximality is closed under supersets.

The semantic constraint breaks this closure. Consider $C = \emptyset$, $P = \{p, u\}$ with $\{p, u\} \in E$, and suppose pivot p is chosen so that u is suppressed as an expansion root. The only sub-tree explored is rooted at p . If $\text{avgSim}(\{p, u\}) < \tau$, this branch is pruned and yields nothing. If $\{u, w\}$ for some $w \in X$ is a valid MSC, it is permanently lost: the sub-tree rooted at u was never entered, and the CHECK pass offers no remedy since it operates only on cliques already constructed, not on suppressed sub-trees. The absence of

pivoting is therefore a correctness requirement, not a design choice. **Complexity.**

THEOREM 3.1 (TIME AND SPACE COMPLEXITY). *Let $\mathcal{L}$ be the feasible terminal states reached by the outer search and let X_C be the candidate set examined by CHECK for $C \in \mathcal{L}$. The running time is $O^*(n2^d + \sum_{C \in \mathcal{L}} 2^{|X_C|})$, where O^* suppresses polynomial set-intersection costs. A graph-independent upper bound is $O^*(3^n)$. With p workers, the implementation uses $O(m + nD + pn^2)$ worst-case space, where D is the vector dimension.*

PROOF. A degeneracy seed has at most d later neighbors. Because no pivot is applied, its outer recursion may visit every subset of those neighbors, giving $O^*(n2^d)$ states. For a feasible terminal C , CHECK may independently visit every subset of X_C . These states are not subsumed by the outer-tree bound because the same earlier extension can be reconsidered for several terminals. Charging a state to a pair of nested vertex sets yields the safe $O^*(3^n)$ global bound. Similarities and sparse adjacency require $O(m + nD)$ storage. Each worker owns an $O(n)$ accumulator, while the decreasing candidate arrays retained along a worst-case CHECK stack sum to $O(n^2)$. $\square$

Design Implication. SEMBK is the direct semantic-search baseline and establishes exact inclusion-wise maximality through its CHECK pass. Its pivot-free branching and repeated extension checks arise from the same property: average similarity couples all pairs in a clique and cannot be summarized by local adjacency alone. Section 4 develops two exact search organizations that expose additional structural or semantic pruning.

4 PROPOSED ALGORITHMS

This section presents two exact algorithms that exploit the global structure of the average constraint from different directions.

STRSUB (Section 4.1) resolves the pivot incompatibility by decoupling structural and semantic search. The pivot is applied without modification to a τ -agnostic structural phase, restoring the $O(3^{n/3})$ BK guarantee; τ is then enforced in a subsequent intra-clique enumeration phase. Complementary sorted-edge and vertex-contribution upper bounds introduced in Phase 2 skip subset-size layers that cannot satisfy τ . Its cost is governed by structural MCE and by the sizes of the resulting host cliques.

MONOSEMMCE (Section 4.2) departs from the BK framework entirely. Motivated by the observation that STRSUB, despite its pruning improvements, still processes all structural maximal cliques irrespective of their semantic quality, MONOSEMMCE introduces a canonical reverse-search strategy in which τ drives enumeration from the first expansion step. A deterministic feasible parent gives each feasible clique one generation path, while depth-first traversal avoids storing a complete size layer. The method is particularly effective at high τ , where few candidate extensions pass the surplus test.

Graph size and structural clique growth determine the crossover between the two methods. We therefore evaluate them on the same quantile grid: STRSUB provides broad threshold coverage, while MONOSEMMCE specializes the search for selective high- τ workloads.

Algorithm 2: STRSUB: Structural Decomposition and Subset Enumeration

Input: $G = (V, E)$, similarities $\{\text{sim}(u, v)\}_{(u,v) \in E}$, threshold τ
Output: All MSCs $\mathcal{R}$

```

1  $\mathcal{F} \leftarrow \emptyset$ ;
  /* Pipelined structural MCE and semantic processing */
2 Build degeneracy ordering  $\sigma$ ;
3 for each  $v \in \sigma$  in parallel do
4    $P \leftarrow \{u \in N(v) \mid \sigma(u) > \sigma(v)\}$ ;
5    $X \leftarrow \{u \in N(v) \mid \sigma(u) < \sigma(v)\}$ ;
6   Run BK( $\{v\}, P, X$ ) with the following callback at each leaf  $M$ ;
  /* Callback: intra-clique semantic enumeration */
7 for each structural leaf  $M$  produced by a worker do
8   if  $\text{avgSim}(M) \geq \tau$  then  $\mathcal{F} \leftarrow \mathcal{F} \cup \{M\}$ ; continue;
  /* Admissible layer bounds */
9   Sort edges of  $M$  by similarity descending;
10   $k^* \leftarrow \max\{k \in [2, |M| - 1] \mid e_k \geq \tau\}$ ;
11  if  $k^*$  does not exist then continue;
12   $lmax \leftarrow \emptyset$ ;  $umask \leftarrow 0$ ;
13  for  $k \leftarrow k^*$  to 2 do
14    Compute the vertex-contribution bound  $\bar{q}_k$  for size  $k$ ;
15    if  $\bar{q}_k < \tau$  then continue;
16    for each size- $k$  subset  $S \subseteq M$  do
17      if  $(S \& umask) = S$  and  $\exists T \in lmax : S \subseteq T$  then
18        continue;
19      if  $\text{avgSim}(S) \geq \tau$  then
20         $lmax \leftarrow lmax \cup \{S\}$ ;
21         $umask \leftarrow umask \mid S$ ;
22         $\mathcal{F} \leftarrow \mathcal{F} \cup \{S\}$ ;
  /* Phase 3: Exact deduplication and strict-containment filtering */
23 Sort  $\mathcal{F}$  by  $|\cdot|$  descending; remove exact duplicates;
24 Initialize an empty inverted index  $inv : v \mapsto \emptyset$ ;
25  $\mathcal{R} \leftarrow \emptyset$ ;
26 for each equal-size group  $\mathcal{G} \subseteq \mathcal{F}$  in descending size do
27   for each  $S \in \mathcal{G}$  in parallel do
28      $r \leftarrow \arg \min_{v \in S} |inv[v]|$ ;
29     Keep  $S$  iff no  $T$  referenced by  $inv[r]$  satisfies  $S \subseteq T$ ;
30 Insert every kept  $S \in \mathcal{G}$  into  $\mathcal{R}$  and  $inv$ ;
31 return  $\mathcal{R}$ ;

```

4.1 STRSUB: Structural Decomposition and Subset Enumeration

Structural–Semantic Decoupling and Containment. The validity of the Bron–Kerbosch pivot rests on a structural invariant: every structural maximal clique must contain at least one vertex from $P \setminus N(u)$ for any pivot $u \in P \cup X$. This invariant is violated when the average-similarity constraint is incorporated directly into the branching procedure, because avgSim is a *global* property of a candidate set and does not decompose along the local neighborhood structure used to justify the pivot. Consequently, the semantic variant SEMBK must forgo pivoting entirely, which undermines the $O(3^{n/3})$ worst-case guarantee of the standard algorithm [44].

Figure 3 separates the two host-clique outcomes. Gray marks structural or global processing, green marks accepted feasible states, and amber marks the bounded search triggered by an infeasible host. Thus M_1 is accepted as C_1 , whereas subset enumeration within M_2 recovers C_2 .

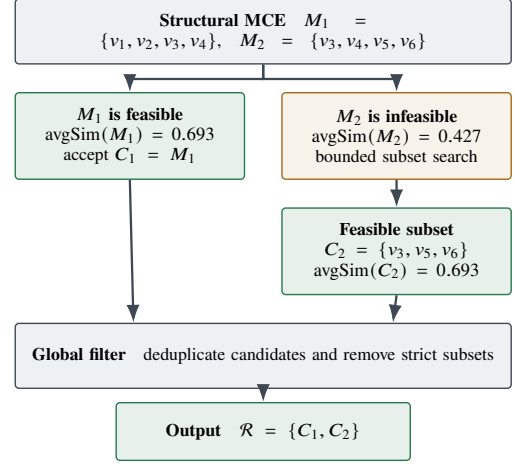


Figure 3: STRSUB execution on the running example.

The key insight behind STRSUB is that the structural and semantic layers of the problem can be *decoupled*. Specifically, the Tomita pivot can be applied without modification to the purely structural problem of enumerating maximal cliques in G , because that problem carries no average-similarity constraint. The threshold τ is then enforced in a second, independent pass confined to each structural maximal clique. The theoretical basis for this decoupling is the following containment lemma.

LEMMA 4.1 (STRUCTURAL SUPERSET OF MSCs). *Every MSC is contained in at least one structural maximal clique of G . That is, for every MSC C , there exists a structural maximal clique M such that $C \subseteq M$.*

PROOF. By definition, every MSC is a clique of G . Let C be an arbitrary MSC, and consider the family

$$\mathcal{F} = \{D \subseteq V \mid C \subseteq D \text{ and } D \text{ is a clique in } G\}.$$

Since $C \in \mathcal{F}$, the family is nonempty. Because V is finite, $\mathcal{F}$ is finite, so we may choose a set $M \in \mathcal{F}$ that is maximal under inclusion. By construction, M is a clique containing C , and no vertex $w \in V \setminus M$ extends M to a larger clique; hence M is a structural maximal clique of G , and $C \subseteq M$. $\square$

Lemma 4.1 establishes that the complete collection of MSCs can be recovered by (i) enumerating all structural maximal cliques, and (ii) searching within each structural maximal clique for semantically feasible maximal subsets. Step (i) is a purely structural problem to which the full Tomita pivot applies, restoring the $O(3^{n/3})$ worst-case bound of standard BK. The threshold τ enters only in step (ii), where the search space is bounded by the size of a single structural clique. Algorithm 2 implements this strategy in three phases.

[Lines 1–6]: Phase 1 – Structural Maximal Clique Enumeration. Phase 1 enumerates all structural maximal cliques of G using the standard BK algorithm equipped with the Tomita pivot [44] and degeneracy ordering [14]. Vertices are processed in degeneracy order σ ; for each seed vertex v , the candidate set P and exclusion set X are initialized to the forward and backward neighborhoods of v in σ , respectively, and recursive BK is invoked on the triple $(\{v\}, P, X)$. Seeds are dispatched in parallel via OpenMP dynamic scheduling. Crucially, this phase is entirely τ -agnostic: the threshold plays no role in branching decisions, so the full pivot strategy and its associated complexity bound are preserved. The implementation does not materialize the set of all structural maximal cliques: when BK reaches a leaf M , the same worker immediately invokes the Phase 2 callback and retains only its thread-local semantic candidates.

[Lines 7–20]: Phase 2 – Intra-Clique Semantic Enumeration. For each structural maximal clique M of size m produced by Phase 1, Phase 2 enumerates all semantically maximal subsets of M with respect to τ . Subsets of M are generated in descending order of cardinality using native-word bitmasks when possible and dynamic bitsets for larger host cliques; no mask-width limit is imposed. Pairwise similarities are precomputed into an $m \times m$ local matrix to amortize repeated avgSim evaluations.

A *local maximality filter* prunes redundant candidates during enumeration. The filter maintains a set $lmax$ of valid subsets discovered so far within M , together with a union mask $umask = \bigcup_{T \in lmax} T$ that provides a constant-time necessary condition for subsumption. A candidate subset S is skipped if $S \subseteq T$ for some $T \in lmax$, verified first via the bitmask test $(S \& umask) = S$ and then by a linear scan over $lmax$. This filter is correct because, if $S \subseteq T$ and $\text{avgSim}(T) \geq \tau$, then S cannot be semantically maximal in G regardless of its own average similarity. Structural leaves are processed by their owning workers. The candidate size layers are further reduced by the admissible bounds described below.

[Lines 21–28]: Phase 3 – Global Deduplication. Because the same MSC may be structurally contained in multiple distinct maximal cliques, Phase 2 may emit duplicate or subsumed candidates across cliques. Phase 3 consolidates these candidates into the final result set $\mathcal{R}$.

The phase proceeds in two steps. First, candidates are sorted by decreasing size with a lexicographic tie break, and exact duplicates are removed in $O(N \log N)$ time, where $N = |\mathcal{F}|$. Second, equal-size groups are processed in that order. The inverted index $inv : v \mapsto \{T \in \mathcal{R} \mid v \in T\}$ contains only retained strict supersets of the current group. For each candidate S , the algorithm identifies its *rarest vertex* $r = \arg \min_{v \in S} |inv[v]|$ and scans only $inv[r]$; a sorted-merge traversal determines in $O(|S| + |T|)$ time whether $S \subseteq T$. Queries within one group read the same frozen index and run in parallel, after which the retained candidates are inserted. Equal-size sets cannot strictly contain one another. Moreover, if a discarded larger candidate contains S , transitivity guarantees that a retained candidate containing both will be indexed. The grouped filter is therefore equivalent to testing all larger candidates while avoiding their indexing and synchronization costs.

Admissible Layer Pruning. Phase 2 as described iterates over all subset sizes from $|M| - 1$ down to 2. A key observation allows the starting size to be reduced: the best achievable avgSim for any

size- k subset of M is bounded above by the mean of the $\binom{k}{2}$ largest pairwise similarities in M .

PROPERTY 4.1 (SORTED-EDGE UPPER BOUND). *Let $e_1 \geq e_2 \geq \dots \geq e_{\binom{m}{2}}$ be the pairwise similarities of M sorted in descending order. For any size- k subset $S \subseteq M$,*

$$\text{avgSim}(S) \leq \bar{e}_k := \frac{1}{\binom{k}{2}} \sum_{i=1}^{\binom{k}{2}} e_i.$$

PROOF. Any size- k subset $S \subseteq M$ uses exactly $\binom{k}{2}$ pairs, each contributing at most the corresponding value in the sorted sequence $e_1 \geq e_2 \geq \dots$; hence $\text{simSum}(S) \leq \sum_{i=1}^{\binom{k}{2}} e_i$, and dividing both sides by $\binom{k}{2}$ gives the result. $\square$

The sorted-edge bound ignores that its largest edges may be concentrated on too few vertices to coexist in one size- k subset. A complementary bound tracks which vertices support those similarities. For each $v \in M$, let $q_v(k)$ be the sum of the $k - 1$ largest similarities between v and vertices in $M \setminus \{v\}$. Let $q_{(1)}(k) \geq \dots \geq q_{(m)}(k)$ denote these values in descending order.

PROPERTY 4.2 (VERTEX-CONTRIBUTION UPPER BOUND). *For any size- k subset $S \subseteq M$,*

$$\text{avgSim}(S) \leq \bar{q}_k := \frac{1}{k(k-1)} \sum_{i=1}^k q_{(i)}(k).$$

PROOF. For every $v \in S$, the sum of its similarities to the other vertices of S uses $k - 1$ vertices from $M \setminus \{v\}$ and is therefore at most $q_v(k)$. Summing over S counts each internal pair twice, so

$$2\text{simSum}(S) \leq \sum_{v \in S} q_v(k) \leq \sum_{i=1}^k q_{(i)}(k).$$

Division by $k(k-1)$ gives the claimed average-similarity bound. $\square$

Note that $\bar{e}_k$ is non-increasing in k : a larger subset requires more edges, drawing further into the sorted list and lowering the mean. Before enumeration, all pairwise similarities of M are sorted once in $O(m^2 \log m)$ and prefix means $\bar{e}_k$ are computed for all k . The algorithm identifies

$$k^* = \max\{k \in [2, |M| - 1] \mid \bar{e}_k \geq \tau\},$$

and begins enumeration at $k = k^*$ rather than $|M| - 1$. Because $\bar{e}_k$ is non-increasing, all sizes $k > k^*$ satisfy $\bar{e}_k < \tau$, certifying that no subset of those sizes can meet the constraint; they are skipped entirely. If no such k^* exists, the whole clique M is skipped. For every remaining size k , the algorithm also skips the layer when $\bar{q}_k < \tau$. The two tests are independently admissible, so using their minimum cannot remove a feasible subset. Computing all per-vertex similarity lists and prefix sums costs $O(m^2 \log m)$, matching the sorting order of the Top-Edge preprocessing.

Characteristics and Applicable Regime. STRSUB inherits the full efficiency of BK with Tomita pivot in Phase 1, which runs in $O(dn \cdot 3^{d/3})$ time independently of τ . At low τ , nearly every subset of a structural maximal clique satisfies the average constraint, so the local maximality filter rapidly identifies large valid subsets and terminates Phase 2 early. The two layer bounds extend this

efficiency to medium and high τ : for most structural cliques M , either the whole clique is accepted immediately ($\text{avgSim}(M) \geq \tau$) or k^* is small, confining bitmask enumeration to a narrow band of sizes near the base. At high selectivity, Phase 1 still enumerates every structural maximal clique. MONOSEMMCE complements this organization by applying τ from the first expansion and storing only depth-first search state.

Complexity.

THEOREM 4.1 (TIME COMPLEXITY OF STRSUB). *STRSUB runs in*

$$O\left(dn \cdot 3^{d/3} + \sum_{M \in \mathcal{M}} 2^{|M|} |M|^2 + KN \log N + K \sum_{S \in \mathcal{F}} L_{\min}(S)\right)$$

time, where d is the degeneracy of G , $N = |\mathcal{F}|$ is the number of candidates before deduplication, $K = \max_{S \in \mathcal{F}} |S|$, and $L_{\min}(S) = \min_{v \in S} |\text{inv}[v]|$.

PROOF. **Phase 1.** With degeneracy ordering, BK with Tomita pivot runs in $O(dn \cdot 3^{d/3})$ time [14].

Phase 2. For a structural maximal clique M of size m , bitmask enumeration visits at most 2^m subsets. Each subset S requires computing $\text{avgSim}(S)$ from the precomputed $m \times m$ similarity matrix at cost $O(|S|^2) \leq O(m^2)$. The per-clique cost is therefore $O(2^m \cdot m^2)$, giving $\sum_{M \in \mathcal{M}} O(2^{|M|} \cdot |M|^2)$ overall. The union-mask pre-filter and local maximality scan do not increase this asymptotic bound.

Phase 3. Lexicographic sorting costs $O(KN \log N)$. For each candidate S , the algorithm scans $\text{inv}[r]$ of length $L_{\min}(S)$; each of the at most $L_{\min}(S)$ subsumption checks costs $O(K)$ via sorted-merge comparison. The total Phase 3 cost is $O(KN \log N + K \sum_{S \in \mathcal{F}} L_{\min}(S))$. $\square$

Remark. The $\sum_{M \in \mathcal{M}} 2^{|M|} \cdot |M|^2$ term is a worst-case bound on Phase 2 without pruning. In practice, the sorted-edge bound (Property 4.1) restricts enumeration to sizes $k \leq k^*$, and the vertex-contribution bound (Property 4.2) may remove additional individual layers, reducing the per-clique cost to $O(|M|^2 \cdot \sum_{k=2}^{k^*} \binom{|M|}{k})$, which is substantially smaller than $2^{|M|} \cdot |M|^2$ whenever k^* is small relative to $|M|$.

4.2 MONOSEMMCE: High- τ Canonical Reverse Search

Motivation and Core Insight. Even with the Top-Edge Layer Pruning, STRSUB is bound to Phase 1: it enumerates all structural maximal cliques before any semantic evaluation takes place. In the high- τ regime, the vast majority of structural cliques may be semantically barren, yet each still contributes Phase 1 cost. More fundamentally, the BK framework offers no mechanism to incorporate τ into branching decisions—the threshold appears only in validation, never in the search logic itself. This structural overhead motivates abandoning the BK skeleton entirely.

The obstacle to integrating τ directly into search is the non-monotonicity of avgSim : an infeasible clique can have a feasible strict superset. A search that retains only feasible states is therefore complete only if every feasible target has a feasible construction path. Theorem 4.2 supplies such a path through the deletion rule defined next.

For a vertex v in a clique C , define its *similarity contribution* as $a_C(v) = \sum_{u \in C \setminus \{v\}} \text{sim}(u, v)$: the sum of its similarities to the other vertices in C . We delete a vertex with minimum similarity contribution, breaking ties by vertex identifier. The theorem below proves that the remaining clique is feasible. This deterministic rule gives every feasible clique a unique parent, called its *canonical parent*. Reverse search accepts an extension only when its parent under this rule is the current clique. Thus every feasible clique is generated once without a visited table or per-layer deduplication.

The implementation traverses each canonical tree depth-first and parallelizes its feasible edge roots. It rejects an extension whenever its updated surplus is negative. Unlike the previous level-synchronous algorithm, it stores no complete size layer; working memory consists of the graph representation, thread-local accumulators, DFS scratch space, and output candidates. A final containment filter is still necessary, because absence of a feasible single extension is weaker than inclusion-wise maximality.

The Monotone Path Property.

THEOREM 4.2 (FEASIBLE MINIMUM-CONTRIBUTION DELETION). *Let C be a τ -semantic clique of size $k \geq 3$, with the similarity contribution $a_C(v)$ defined above. If $d(C)$ minimizes the pair $(a_C(v), v)$ lexicographically, then $C \setminus \{d(C)\}$ is a τ -semantic clique. Repeated application of this rule therefore reaches a feasible edge, and reversing these deletions gives a non-increasing-average insertion path.*

PROOF. Each pair is counted once at each endpoint. The minimum contribution therefore satisfies $a_C(d(C)) \leq 2\text{simSum}(C)/k$. Deleting $d(C)$ leaves similarity sum at least $\text{simSum}(C)(k-2)/k$, and hence

$$\text{avgSim}(C \setminus \{d(C)\}) \geq \frac{\text{simSum}(C)(k-2)/k}{\binom{k-1}{2}} = \text{avgSim}(C) \geq \tau.$$

Deterministic tie-breaking makes $d(C)$ unique. Applying the same argument until two vertices remain preserves feasibility at every deletion; reversing the sequence proves the insertion-path claim. $\square$

Figure 4 applies this rule to the running example. The green paths insert the canonical deletion of each child, while the red path reaches C_2 through a non-owning insertion and stops before recursion.

Feasible Edge Roots. Every feasible clique of size at least two reaches a feasible edge under the deterministic deletion rule, so Algorithm 3 starts one tree for each edge with similarity at least τ . A state stores the clique C , its pairwise sum S , the common-neighbor set P , candidate accumulators acc , and the member contributions a_C . No BK exclusion set is required: uniqueness follows from the canonical-parent test rather than from an insertion order.

Surplus and Canonical-Child Tests. For $k = |C|$ and $\delta = \text{acc}[v]$, the extension satisfies

$$W(C \cup \{v\}) = W(C) + \delta - \tau k.$$

If this value is negative, the child itself is infeasible and cannot lie on the canonical path of any feasible answer, so it is safely skipped. For a feasible child $D = C \cup \{v\}$, the algorithm updates every member's similarity contribution and accepts D only if $v = d(D)$. Equivalently, the canonical parent of D must be exactly C . This test

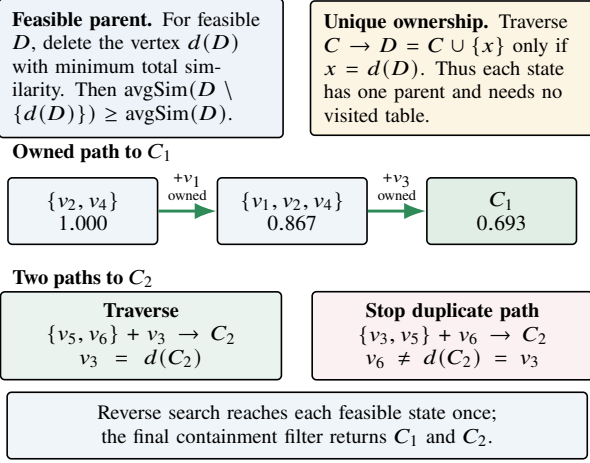


Figure 4: Canonical ownership on the running example.

Algorithm 3: MONOSEMMCE: Canonical Reverse Search

```

1 Input:  $G = (V, E)$ , semantic vectors  $\{x_v\}$ , threshold  $\tau$ 
2 Output: All MSCs  $\mathcal{R}$ 
3 Precompute  $\text{sim}(u, v)$  for all  $(u, v) \in E$ ;
4  $\mathcal{F} \leftarrow \emptyset$ ;
5 for each  $(u, v) \in E$  with  $u < v$  and  $\text{sim}(u, v) \geq \tau$  in parallel do
6    $C \leftarrow \{u, v\}$ ;  $S \leftarrow \text{sim}(u, v)$ ;
7    $P \leftarrow N(u) \cap N(v)$ ;
8    $\text{acc}[w] \leftarrow \text{sim}(w, u) + \text{sim}(w, v)$  for  $w \in P$ ;
9    $a_C(u) = a_C(v) \leftarrow \text{sim}(u, v)$ ;
10  Search( $C, S, P, \text{acc}, a_C$ );
11 Apply Algorithm 2's grouped strict-containment filter to  $\mathcal{F}$ ,
    obtaining  $\mathcal{R}$ ;
12 return  $\mathcal{R}$ ;
13 Function Search( $C, S, P, \text{acc}, a_C$ ):
14    $k \leftarrow |C|$ ;  $W \leftarrow S - \tau \binom{k}{2}$ ;
15    $\text{hasValid} \leftarrow \text{false}$ ;
16   for each  $v \in P$  do
17      $\delta \leftarrow \text{acc}[v]$ ;
18     if  $W + \delta - \tau k < 0$  then continue;
19      $\text{hasValid} \leftarrow \text{true}$ ;
20      $D \leftarrow C \cup \{v\}$ ;
21     Update  $a_D$  from  $a_C$  and  $\text{sim}(v, u)$  for  $u \in C$ ;
22     if  $v \neq \arg \min_{x \in D} (a_D(x), x)$  then continue;
23      $P' \leftarrow P \cap N(v)$ ;
24      $\text{acc}'[w] \leftarrow \text{acc}[w] + \text{sim}(w, v)$  for  $w \in P'$ ;
25     Search( $D, S + \delta, P', \text{acc}', a_D$ );
26   if not  $\text{hasValid}$  then  $\mathcal{F} \leftarrow \mathcal{F} \cup \{C\}$ ;
  
```

removes alternative insertion paths before recursion rather than after state materialization.

The path property is used to assign unique parent ownership rather than to order candidates. The algorithm tests child feasibility directly and then applies the minimum-contribution deletion rule.

This removes duplicate paths before recursion without a candidate-order cutoff. Accumulator values are restored from saved values when backtracking, so the result does not depend on reversing a sequence of floating-point additions.

Inclusion-Wise Maximality. A state with no feasible single-vertex extension is added to $\mathcal{F}$, but is not yet declared an MSC. If such a candidate C is not inclusion-wise maximal, the finite family of feasible clique supersets of C contains an inclusion-wise maximal member M . The canonical search enumerates M exactly once, and M has no feasible single extension, so it also belongs to $\mathcal{F}$. The final grouped inverted-index containment filter consequently removes C . Conversely, an MSC is not contained in any other feasible candidate and is retained.

THEOREM 4.3 (ENUMERATION CORRECTNESS). *Algorithm 3 generates every τ -semantic clique of size at least two exactly once and returns exactly all MSCs.*

PROOF. Existence follows by reversing the repeated deterministic deletions in Theorem 4.2; every prefix is feasible and therefore passes the surplus test. Uniqueness follows because every non-root feasible clique has the single parent $C \setminus \{d(C)\}$, and only that parent accepts it as a child. The preceding containment argument proves that the final filter retains precisely the inclusion-wise maximal feasible cliques. $\square$

Complexity.

THEOREM 4.4 (TIME AND SPACE COMPLEXITY OF MONOSEMMCE). *Let F_τ be the number of τ -semantic cliques of size at least two, K the maximum clique size, Δ the maximum degree, T the number of worker threads, and $N = |\mathcal{F}|$ the number of terminal candidates before containment filtering. A conservative bound for canonical search is*

$$O\left(F_\tau \Delta K \log \Delta + KN \log N + K \sum_{S \in \mathcal{F}} L_{\min}(S)\right),$$

and its space usage is $O(m + T(n + K\Delta) + NK)$ words, in addition to vector storage.

PROOF. Canonical uniqueness bounds accepted search states by F_τ . At one state, at most Δ candidates are tested. Updating and comparing the $O(K)$ similarity contributions uses cached edge similarities, each found in $O(\log \Delta)$ time; candidate intersection and accumulator updates fit within the stated conservative per-state bound. The inverted-index filter sorts N candidates and performs at most $L_{\min}(S)$ exact $O(K)$ subset checks for each S .

Similarity adjacency lists and sparse bitmaps use $O(m)$ words. Each worker owns an $O(n)$ generation-tagged accumulator and at most $O(K\Delta)$ DFS scratch storage. Crucially, this term depends on search depth rather than the number of feasible cliques at any size. The terminal candidates occupy $O(NK)$ words. $\square$

Remark. The canonical tree contracts as τ reduces the feasible-state family F_τ , giving MONOSEMMCE its high-threshold operating region. STRSUB provides the complementary structural decomposition across broader thresholds.

5 IMPLEMENTATION OPTIMIZATIONS

The following optimizations are applied to MONOSEMMCE and STRSUB without affecting correctness. They address three categories of bottlenecks: parallel load balance, repeated computation in the expansion inner loop, and memory access cost for neighbor intersection and similarity lookup.

Parallel Decomposition and Load Balance. STRSUB Phase 1 parallelizes BK over seed vertices under degeneracy ordering: each seed v spawns an independent subproblem over its forward neighborhood (P_v, X_v) , and distinct seeds share no mutable state and require no synchronization. Phase 2 normally processes each structural maximal clique with thread-local candidate lists, but a few large host cliques can dominate this work. For such a clique, each fixed-cardinality layer is partitioned by its smallest selected vertex. These partitions are disjoint and may run concurrently because distinct equal-size subsets cannot strictly contain one another. A barrier merges the feasible subsets before the next smaller layer, preserving all local-containment decisions while exposing parallelism inside a single expensive host clique. Thread-local candidates are merged only after processing, avoiding lock contention on the output.

MONOSEMMCE assigns each feasible edge root to one canonical tree. Distinct trees contain disjoint feasible-clique states, so roots are dispatched with dynamic OpenMP scheduling and require no shared visited table. Each worker performs DFS with thread-local accumulators, scratch buffers, statistics, and candidate output; only the final merge and containment filter synchronize globally.

Incremental Candidate Maintenance. When expanding state $(C, \text{simSum}, P, \text{acc})$ by vertex v , the child state requires an updated candidate set $P' = P \cap N(v)$ and an updated accumulator acc' . A naïve implementation recomputes acc' from scratch at cost $O(|P'| \cdot |C|)$ per expansion.

Since $\text{acc}[w] = \sum_{u \in C} \text{sim}(w, u)$, the child accumulator satisfies

$$\text{acc}'[w] = \text{acc}[w] + \text{sim}(w, v) \quad \forall w \in P',$$

reducing the update to $O(|P'|)$ regardless of clique depth. The DFS records each previous accumulator value and restores it exactly after the recursive call. Reversing an update by subtracting the same floating-point similarity is not equivalent: rounding can accumulate across sibling branches and change a threshold decision. Candidate ordering has no correctness role in canonical search, so the standard implementation performs no per-state sort.

Sparse Blocked Bitmap for Neighbor Intersection. Candidate set intersection $P' = P \cap N(v)$ is the innermost operation of every expansion. We represent each adjacency list as a *sparse blocked bitmap*: vertices are indexed by $\lfloor v/64 \rfloor$, and only non-empty 64-bit blocks are stored as (index, word) pairs in sorted order. Intersection of two rows executes a merge-style scan over their block lists, applying bitwise AND only to matching block indices and extracting set bits via `ctz`. The cost is $O(b_u + b_v)$ where b_u, b_v count non-empty blocks in each row, compared to $O(n/64)$ for a dense bitmap.

This advantage compounds with search depth: at layer ℓ , every candidate $w \in P$ is a common neighbor of all $\ell+2$ clique members, so $|P|$ decreases with ℓ and each row’s active block count shrinks accordingly. The per-intersection cost therefore decreases with depth.

Similarity Computation with Reorder-Aware Caching. All pairwise similarities are precomputed once and stored in sorted adjacency lists for $O(\log \deg(u))$ lookup via binary search. Inner products use 8-wide AVX2 SIMD over the float32 vector store. Edge similarities remain float32 to control memory traffic, while similarity sums, per-candidate accumulators, surpluses, and threshold products are maintained in double precision. Every feasibility test compares these double-precision quantities directly with an inclusive boundary; no algorithm-specific epsilon is used.

To reduce cache miss cost on repeated lookups, vertices are reordered by a degree-descending BFS before vectors are stored contiguously in memory. A vertex u with degree d_u participates in $O(d_u)$ expansions as a candidate across the search; placing high-degree vertices at low memory offsets concentrates the most frequently accessed vectors into a small number of cache lines, reducing cold misses on the hot working set.

6 EXPERIMENTS

6.1 Experimental Setup

Algorithms. We evaluate four implementations derived from one shared graph and vector backend. STRUCTBK is a Tomita-pivot structural MCE implementation and serves as the single structural reference; it does not solve MSC enumeration. SEMBK is the pivot-free semantic baseline of Section 3. STRSUB and MONOSEMMCE are the exact algorithms of Section 4. Runtime experiments use instrumentation-free binaries. Search counters use separately compiled statistics modes and never contribute timing values.

Datasets and attributes. Table 1 fixes all dataset roles before comparison. The common grid contains three graphs with natural semantic vectors and five public graphs with controlled vectors. FB15K-237 and WN18RR entity descriptions are encoded by all-MiniLM-L6-v2; ogbn-arxiv retains its official 128-dimensional node features [20, 45]. For the public structural graphs from SNAP, SuiteSparse, and the Enron communication collection [11, 26], PCG64 draws 256-dimensional coordinates independently from $U[-1, 1)$ using a fixed dataset seed, after which the shared loader applies ℓ_2 normalization. Thus only attributes are generated; vertices and edges remain those of the public graph. The same graph/vector provenance rule applies to every generated attribute row in the table.

For the scale experiment, vertices of ogbn-arxiv are ordered by decreasing source-graph degree with original identifier as the tie break. Each level is the full induced graph on an order prefix and copies the corresponding official feature rows. The provenance record stores the construction rule, identifier mapping, and source/output checksums. Every level retains all induced edges and the corresponding official feature rows.

Environment. Experiments run on an AMD Ryzen 9 7945HX workstation with 16 physical cores, 32 hardware threads, and 32 GiB memory under Windows 11 and MSYS2 GCC 13.2. Every process has a 16 GiB hard memory limit and uses at most 32 OpenMP threads. Following the common use of censored outcomes in large-graph mining evaluations [28, 36], the formal evaluation harness

Dataset	$ V $	$ E $	Dim.	Evaluation role
FB15K-237	14,541	248,611	384	natural, common grid
WN18RR	40,943	75,762	384	natural, common grid
ogbn-arxiv	169,343	1,157,799	128	natural, common grid
email-Enron	36,692	183,831	256	generated, common grid
sc-nasasrb	54,870	1,311,227	256	generated, common grid
sc-pkustk11	87,804	2,565,054	256	generated, common grid
sc-pwtk	217,891	5,653,221	256	generated, common grid
sc-lldoor	909,537	20,770,807	256	generated, common grid

Table 1: Datasets, vector dimensions, and evaluation roles.

applies a uniform 1,200-second limit and records timeout and out-of-memory outcomes without a result count. Completed configurations are run three times and reported by median algorithm time; a censored configuration is run once because repetition cannot change its status under the fixed limit.

Threshold protocol. Absolute cosine values have different meanings across vector sources. We therefore pre-register per-dataset thresholds as quantiles of cosine similarity over existing graph edges. The common grid uses q_{20} , q_{80} , and q_{99} ; the sensitivity grid adds q_5 , q_{50} , q_{70} , q_{90} , q_{95} , and q_{97} . This grid includes low, intermediate, transition, and highly selective regimes without choosing a dataset-specific point after observing runtime. All methods receive the same stored float32 edge similarities and use double-precision sums with an inclusive feasibility boundary. On sc-lldoor, MONOSEMMCE reaches the 1,200-second limit at q_{20} ; lowering τ only enlarges its feasible root and canonical-state families, so q_5 is censored as TLE without a redundant execution.

6.2 Exactness Before Performance

The vertex-contribution bound matched exact size-layer optima on 225,000 oracle instances and never underestimated an attainable average. Canonical reverse search matched exhaustive feasible-state ownership and inclusion-maximal output on 40,001 oracle instances. The suite covers negative similarities, equality at τ , tied vertex contributions, and feasible supersets unreachable through some insertion orders.

Real-data checks compare normalized clique sets and per-size histograms rather than aggregate counts alone. All three semantic methods return the same 577 MSCs on WN18RR at $\tau = 0.8$ and the same 226,936 MSCs on ogbn-arxiv at q_{80} . STRSUB and MONOSEMMCE agree on all 7,476 FB15K-237 outputs at q_{99} as complete normalized sets. Their aggregate counts and every size-layer count also agree on all jointly completed rows in Table 2, including 13,473,317 sc-lldoor outputs at q_{80} . Each STRSUB ablation preserves the full 76,106-clique FB15K-237 result. These checks cover the final strict-containment filter required by Definition 5, not only search-tree ownership.

6.3 Overall Efficiency

Table 2 is the first end-to-end overview: it applies the same three quantiles to all eight datasets in Table 1, thereby exposing low-, intermediate-, and high-selectivity operating patterns. STRSUB completes 22 of 24 cells and every q_{20} row, confirming that structural

decomposition provides broad low-threshold coverage. At q_{80} , the ranking depends on graph structure: STRSUB leads on FB15K-237 and ogbn-arxiv, while MONOSEMMCE leads on the other six datasets. At q_{99} both methods complete every dataset, and canonical search is $2.17\times$ – $6.24\times$ faster. Thus the high-threshold specialization is visible across natural and generated attributes rather than only on one extreme instance.

Runtime cells report seconds. Green marks the fastest completed MSC method, TLE denotes the uniform 1,200-second limit, and STRUCTBK reports only the threshold-independent structural enumeration time.

SEMBK provides the direct semantic-search reference in the same grid. All three methods complete WN18RR within 0.023–0.066 seconds. On ogbn-arxiv, the fastest specialized method improves over SEMBK by $1.38\times$ – $6.69\times$; on email-Enron, the improvement is $70.8\times$ – $291.2\times$. SEMBK reaches TLE throughout the FB15K-237 and generated-attribute rows except email-Enron. STRUCTBK quantifies the structural enumeration cost independently of τ ; on ogbn-arxiv it enumerates 1,041,789 structural maximal cliques in 0.309 seconds.

6.4 Threshold and Scale Regimes

Figure 5 adds the transition quantiles to two representative regimes. On ogbn-arxiv, STRSUB leads through q_{90} , whereas MONOSEMMCE becomes $2.17\times$ faster at q_{95} and $4.13\times$ faster at q_{97} . The crossover therefore appears before the extreme q_{99} point. The sc-lldoor panel uses the same nine-point grid and shows a wider transition: STRSUB completes the low endpoint, while MONOSEMMCE completes the selective range from q_{70} through q_{99} . The two panels show that the crossover location depends on both semantic selectivity and structural workload.

Figure 6 jointly varies threshold and graph size on nested ogbn-arxiv induced subgraphs. Positive cells indicate that STRSUB is faster; negative cells indicate that MONOSEMMCE is faster. The crossover remains at q_{95} from 25K vertices through the full 169,343-vertex graph. Thus high semantic selectivity, rather than a single favorable graph size, drives the canonical-search advantage. At q_{99} the full graph takes 0.707 seconds for STRSUB and 0.200 seconds for MONOSEMMCE while both return 10,261 MSCs.

6.5 Parallelism

Figure 7 compares the parallel behavior of both specialized algorithms. Dynamic edge-root scheduling gives MONOSEMMCE $5.62\times$ speedup at 32 threads on ogbn-arxiv q_{50} . For STRSUB, statistics collected separately from timing attribute 98.88% of the 32-thread

Dataset	Quantile	MSC count	STRUCTBK (structure only)	SEMBK	STRSUB	MONOSEMCE
FB15K-237	q_{20}	206,470	0.103	TLE	0.221	TLE
	q_{80}	76,106		TLE	9.132	TLE
	q_{99}	7,476		TLE	45.234	10.091
WN18RR	q_{20}	55,200	0.015	0.037	0.058	0.066
	q_{80}	14,379		0.031	0.048	0.036
	q_{99}	750		0.035	0.050	0.023
ogbn-arxiv	q_{20}	925,239	0.309	2.783	1.233	3.357
	q_{80}	226,936		1.206	0.872	1.610
	q_{99}	10,261		1.318	0.843	0.197
email-Enron	q_{20}	223,703	0.090	18.330	0.259	16.739
	q_{80}	60,352		21.369	1.266	0.083
	q_{99}	1,819		9.608	0.206	0.033
sc-nasasrb	q_{20}	28,154	0.090	TLE	0.301	TLE
	q_{80}	834,914		TLE	102.820	0.759
	q_{99}	12,742		TLE	0.415	0.079
sc-pkustk11	q_{20}	22,612	0.143	TLE	0.645	TLE
	q_{80}	2,214,411		TLE	TLE	2.219
	q_{99}	24,891		TLE	0.611	0.167
sc-pwtk	q_{20}	58,523	0.286	TLE	1.093	TLE
	q_{80}	4,483,887		TLE	TLE	3.977
	q_{99}	54,907		TLE	1.364	0.421
sc-lldoor	q_{20}	274,585	1.679	TLE	4.870	TLE
	q_{80}	13,473,317		TLE	563.996	14.944
	q_{99}	202,030		TLE	4.845	1.658

Table 2: Median runtime on the fixed common grid (seconds).

BK-plus-local-processing work to intra-clique subset enumeration on FB15K-237 q_{80} ; the final global filter takes 65 ms. We therefore partition sufficiently large fixed-size subset layers by their smallest selected vertex. Partitions within a layer are independent because equal-size sets cannot strictly contain one another, while a barrier between descending size layers preserves local-containment pruning. This bounded fine-grained schedule reaches 2.39 $\times$ speedup at 16 threads and 2.26 $\times$ at 32 threads without changing the output set or size histogram; the remaining saturation follows the uneven cost of host-clique subset tasks, not final containment filtering.

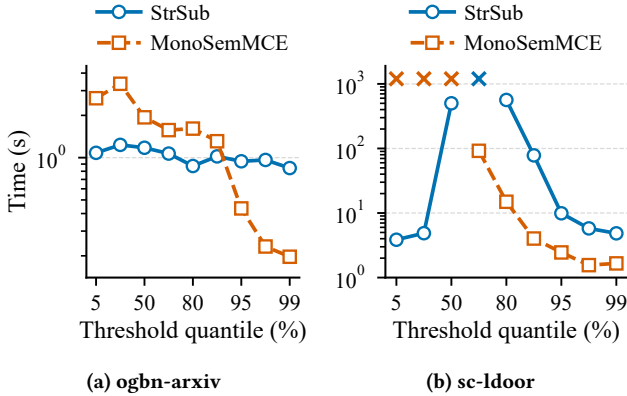


Figure 5: Runtime over the fixed nine-quantile grid.

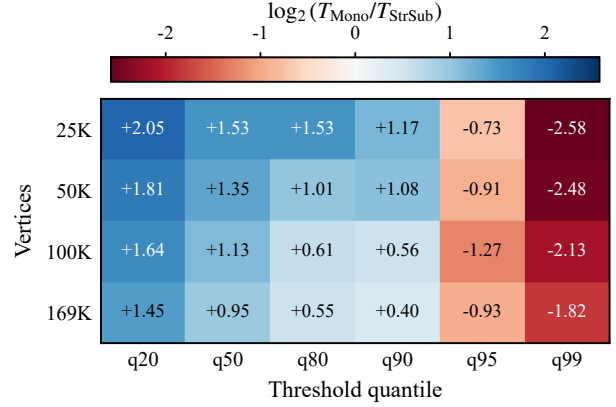


Figure 6: Runtime regimes on nested ogbn-arxiv subgraphs.

6.6 Mechanisms and Exact Ablations

Figure 8 links observed runtimes to algorithm-specific work. On sc-nasasrb at q_{99} , the Top-Edge bound reduces subset tests from 16.50 million to 12.95 million, and the vertex-contribution bound reduces them from 80.28 million to 12.95 million. On ogbn-arxiv, increasing the threshold from q_5 to q_{99} contracts accepted canonical states from 17.08 million to 18,011; at q_{99} , 10,272 terminal candidates are reduced to 10,261 final MSCs. The measurements explain why STRSUB remains stable across much of the grid and why MONOSEMCE changes regime sharply near q_{95} .

Table 3 evaluates the three independently removable STRSUB components on two workloads. All four variants preserve 76,106

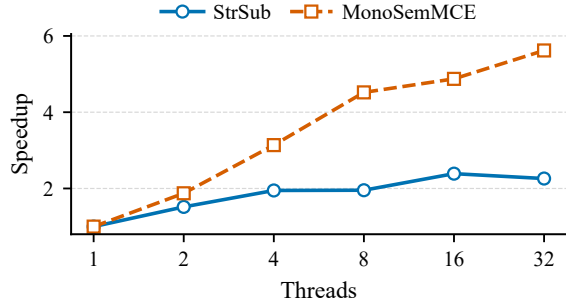


Figure 7: Parallel speedup from 1 to 32 threads.

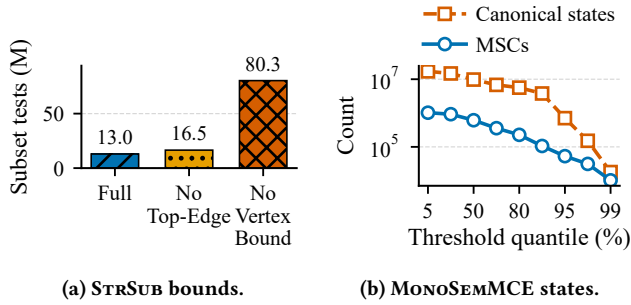


Figure 8: Search counters by algorithm.

Variant	FB15K-237 q_{80}	sc-nasasrb q_{99}
	Time (s) / peak GiB	
Full	8.060 / 0.12	0.287 / 0.24
No Top-Edge	7.462 / 0.12	0.324 / 0.24
No vertex bound	5.898 / 0.12	0.312 / 0.24
No local containment	11.685 / 2.26	0.294 / 0.24

Table 3: STRSUB component ablation (seconds / peak GiB).

MSCs on FB15K-237 and 12,742 on sc-nasasrb. On sc-nasasrb, the complete design is the fastest; the Top-Edge and vertex-contribution bounds reduce subset tests from 16.50 and 80.28 million, respectively, to 12.95 million. On FB15K-237, local containment is the principal materialization control: disabling it increases peak memory from 0.12 to 2.26 GiB and runtime from 8.060 to 11.685 seconds.

Table 4 compares canonical reverse search with two alternative duplicate-control organizations that share the same similarity cache, feasibility test, and final containment filter. Canonical search is 6.27 $\times$ and 4.21 $\times$ faster than visited DFS and layered generation on ogbn-arxiv, and 4.21 $\times$ and 5.38 $\times$ faster on sc-nasasrb. It also has the smallest peak memory in both workloads. The three organizations return identical normalized clique sets and size histograms.

6.7 Output Space and Result Semantics

The final evaluation examines how the threshold changes the returned pattern space and whether the patterns align with external semantic labels. We first measure exact output distributions on FB15K-237, ogbn-arxiv, sc-nasasrb, and sc-lldoor over the fixed q_{20} ,

	ogbn-arxiv q_{95}		sc-nasasrb q_{80}	
Generation	Time (s)	GiB	Time (s)	GiB
Canonical	0.388	0.39	0.729	0.22
Visited DFS	2.432	0.53	3.069	0.64
Layered	1.635	0.49	3.922	0.81

Table 4: MONOSEMMCE generation-strategy ablation.

q_{50} , q_{80} , q_{95} , and q_{99} grid. For each dataset, both the MSC count and mean size are normalized by the structural maximal cliques returned by STRUCTBK on the same graph.

Figure 9 compares the normalized transitions. The output-count ratios aggregated by the geometric mean are 1.04, 7.50, 3.53, 0.61, and 0.11. Intermediate thresholds split large structural cliques into multiple inclusion-maximal feasible subsets, while high thresholds contract the result space. The geometric-mean size ratio decreases from 1.06 at q_{20} to 0.39 at q_{99} . The individual curves retain the distinction between the two natural-semantic graphs and the two generated-attribute graphs, while the aggregate identifies the threshold-wide distribution shift.

The subject-label analysis uses ogbn-arxiv because its 40 official categories are independent of the mining objective. We additionally evaluate PAIRSIM-MCE, which retains an original edge when its end-point similarity reaches τ and then enumerates structural maximal cliques. Subject purity is the largest official-category fraction in an output, averaged over non-singleton outputs. Figure 10 reports each threshold as a joint purity and vertex-coverage operating point; the distribution analysis above supplies the corresponding change in pattern size. Across q_{50} – q_{99} , MSC attains both higher mean purity and higher coverage than PAIRSIM-MCE. STRUCTBK provides the threshold-independent structural reference at full coverage.

Figure 11 visualizes the largest label-pure MSC selected deterministically at q_{80} . The 18 papers all belong to cs.IT and study private information retrieval, spanning coded databases, cache-aided retrieval, private side information, wiretap channels, and private set intersection. Banawan, Ulukus, Wei, Jafar, and Sun recur across the group. Paper positions are initialized by classical multidimensional

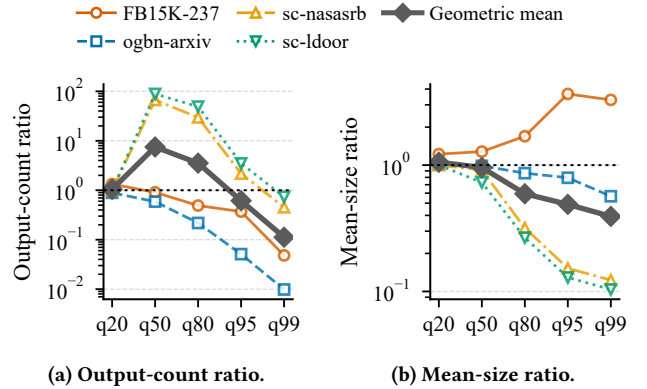


Figure 9: MSC output distributions relative to STRUCTBK.

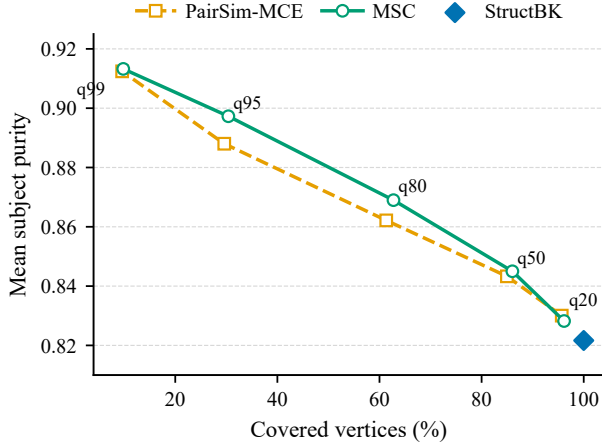


Figure 10: Subject purity and vertex coverage on ogbn-arxiv.

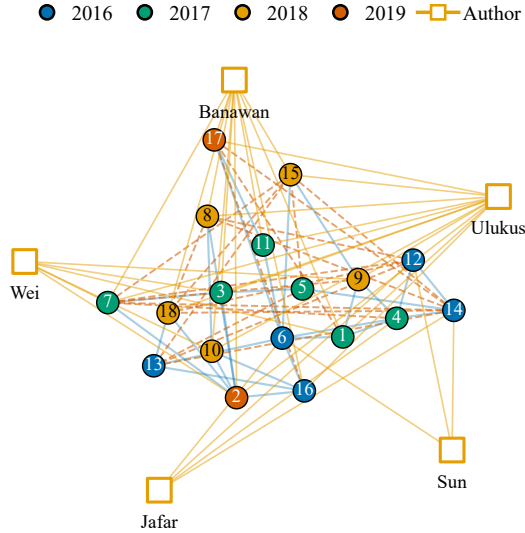


Figure 11: An 18-paper MSC from ogbn-arxiv.

scaling of the cosine distances and relaxed only to separate overlapping nodes. Blue links form the union of each paper’s two most similar neighbors, red dashed links show all below-threshold pairs, and gold links connect authors appearing in at least three papers.

The group has average similarity 0.919 at $\tau = 0.889$, while 20 of its 153 pairs fall below the threshold and its minimum pair similarity is 0.821. The average constraint therefore retains the complete, collectively coherent group that the every-pair threshold divides. The case connects the aggregate label measurements to a concrete collaboration and research topic.

Across these experiments, STRSUB and MONOSEMMCE exhibit overlapping but distinct regions. STRSUB is preferred when structural MCE remains compact and semantic selectivity is low or intermediate. MONOSEMMCE is specialized for thresholds high enough to contract the feasible canonical tree, with the crossover appearing at q_{95} on the controlled scale series. The common grid, transition sweep, and controlled scale series connect this crossover to the two search organizations.

7 RELATED WORK

Maximal Clique Enumeration. The combinatorial output bound for maximal cliques was established by Moon and Moser [32], and polynomial-delay enumeration developed through maximal independent sets [30, 46] and cliques [5]. Bron–Kerbosch backtracking [4] became the standard practical skeleton, with Tomita pivoting attaining the worst-case output order [44]. Degeneracy-oriented variants exploit sparse real-world graphs [7, 13, 14]; out-of-core and parallel systems distribute the resulting search across massive graphs and multicore processors [8, 10, 27, 40]. These techniques justify the single STRUCTBK reference and the structural phase of STRSUB, but their pivot argument depends only on adjacency. It does not transfer to a global non-monotone average constraint.

Attributed Subgraphs. Classical cohesive-subgraph models trade structural strictness for scalable community objectives, including cores, modularity, and influence [6, 16, 29, 34, 41]. Attributed graph clustering combines topology with attribute distance or learned cluster objectives [3, 9, 37, 48, 49]. Pattern-mining formulations more closely connect structure and attributes: CoPaM enumerates dense connected subgraphs homogeneous in a feature subspace [33], GAMER joins subspace clustering with dense-subgraph mining [18], and structural-correlation patterns associate attribute sets with quasi-cliques [42]. Query-oriented systems instead return communities containing specified nodes or attributes under core or truss constraints [15, 21]. MSC differs at the contract level: it enumerates all structurally complete, inclusion-wise maximal sets under one group-average vector constraint. Consequently, these methods motivate result quality comparisons, but are not interchangeable efficiency baselines.

Vector Attributes and Similarity. Vector-attributed graphs arise from knowledge-graph embeddings [2, 12, 35, 43], random-walk representations [17, 38], and message-passing models [19, 25]. Our natural workloads use text-derived sentence representations [39, 47] and the Open Graph Benchmark features [20], with FB15K-237 following the filtered knowledge-base split [45]. Approximate vector search accelerates unconstrained nearest-neighbor retrieval through hashing or navigable graphs [1, 22, 31], and GPU indexes extend this approach to billion-scale collections [23]. MSC needs similarities only on graph edges and must preserve an inclusive threshold exactly, so the implementation computes those edge similarities once and does not introduce approximate-neighbor error into enumeration.

Similarity Search in Vector Spaces. The precomputation of pairwise similarities leverages extensive work on vector similarity search [23]. We employ exact cosine similarity via AVX2 SIMD instructions because similarities are needed only for graph edges,

rather than for all $O(n^2)$ pairs. High-dimensional workloads could instead use locality-sensitive hashing [1] or graph-based nearest-neighbor indexes [31], although approximate integration is beyond the present work.

8 CONCLUSION

We introduced maximal semantic clique (MSC) enumeration under an average pairwise similarity threshold, which identifies structurally complete cliques with guaranteed average pairwise semantic similarity. The problem generalizes classical MCE and is NP-complete. Its defining challenge is the non-monotonicity of the average-similarity constraint over the subset lattice, which invalidates the structural invariants that underpin the efficiency of standard Bron–Kerbosch algorithms.

We used SEMBK as a direct pivot-free baseline and proposed two exact algorithms with distinct operating regions. STRSUB decouples structural and semantic search, restores the Tomita pivot in a τ -agnostic structural phase, and applies admissible edge and vertex-contribution bounds during exact subset enumeration. MONOSEM-MCE is a high- τ specialization: the deterministic deletion rule assigns each feasible clique one parent, so reverse DFS eliminates duplicate insertion paths without visited sets or materialized layers. Both methods use global strict-containment filtering to implement inclusion-wise maximality.

Exact-output tests validate the canonical search and every optional STRSUB component before timing. On ogbn-arxiv, STRSUB is fastest at the intermediate threshold, while MONOSEM-MCE is 4.28 $\times$ faster at q_{99} ; on FB15K-237, the corresponding high-threshold advantage is 4.48 $\times$. Across the eight-dataset common grid, STRSUB completes 22 of 24 cells, and canonical search is 2.17 $\times$ –6.24 $\times$ faster on every q_{99} row. The controlled scale series places the crossover at q_{95} across four graph sizes, while canonical edge-root scheduling reaches 5.62 $\times$ speedup with 32 threads. Mechanism counters connect these results to exact layer bounds in STRSUB and the rapid high-threshold contraction of the canonical tree in MONOSEM-MCE. The fixed five-threshold output study also captures the intermediate pattern-count expansion and high-threshold contraction, while external labels and a real paper group establish the semantic content of the returned MSCs. Future directions include stronger admissible multi-step bounds, finer canonical-subtree scheduling, and dynamic-graph extensions.

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